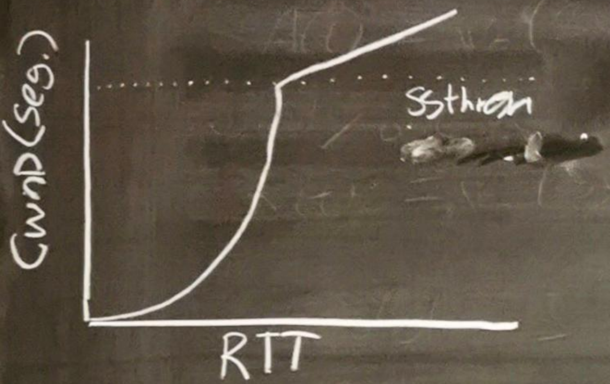
⑤ End2END Performance Using TCP

Congestion Control:



— Maximizing throughput while avoiding Congestion

— Start → cwnd = 1 segment

— Slow Start, exp. ramp up
cwnd·2 after each RTT

— after reaching ssthresh cwnd ↑ 1
per RTT

— ssthresh init → user defined

— Triple dup ACK → cwnd/2 = ssthresh, Start = cwnd/2
(Single Packet Missing ACK) Fast Transmit

— TimeOut → cwnd = 1, ssthresh = cwnd/2
(no Ack before RTO) retransmission TO

— Sliding window → Protect Receiver
Congestion window → Protect network (Routers)
usable window = Min(Slidewin, cwnd)

— Layer 2 can hide Loss due to retransmissions (L2)

— TCP back off may never trigger — cwnd grow
until hit L2 Ceil